

# 1 Planning Variants

## Overview

|                        |                                                      |
|------------------------|------------------------------------------------------|
| Menu                   | Master data → Production control → Planning variants |
| Transaction code       | plvar                                                |
| Function authorization | mdplvar                                              |

## Purpose

Use this function to create or modify planning variants in the system.

## Integration

A planning variant combines specific settings that are relevant for the planning. The settings included in a planning variant are usually used by the function *Automatic assignment*.

## Requirements

The [Priority rule](#) used is defined in the system.

## Selection criteria

The application provides the following selection criteria:

### Planning variant

Enter the required search criterion to search for a planning variant. Please note case sensitivity.  
You can also use wildcards.

### Personal

This checkbox is a "tri-state" checkbox that can have three different states:

- ☐ Only those planning variants are selected that are not identified as *personal*.
- ☒ Only those planning variants are selected that are identified as *personal*.
- ☐ All planning variants are selected.

## Field descriptions

### Planning variant

Name of the planning variant

### Comment

Comment

**Owner**

User name of the owner of the variant.

**Personal**

If a planning variant is identified as personal, only the person who created this variant can use or edit it.

**Planning lead time**

Period of time that the planning function identifies as lead time; result: no assignment is made. The period of time specified is the minimum time that is required to implement a planning.

**Planning time fence**

After an automatic scheduling run, the system automatically fixes the operations that start in the specified planning time fence if the relevant option is enabled in the [HLS settings](#). This means that the planning time fence must not be larger than the planning horizon.

**Priority rule**

You can select a strategy for the automatic planning. You can select from preconfigured rules (HLS-RBM) and self-configured rules (HLS-VMB; HLS-ZMB). The following preconfigured rules are available:

- Lowest setup costs: to calculate the setup costs, setup time + teardown time are totaled (dynamic setup times are not integrated).
- Shortest operation time: operation time = setup time + processing time + teardown time
- Shortest remaining processing time: the remaining processing time is calculated using the remaining run time formula stored for the OP.
- Longest operation time: operation time = setup time + processing time + teardown time
- Longest remaining processing time: the remaining processing time is calculated using the remaining run time formula stored for the OP.

**Displacement to the left**

In case of an automatic planning of operations, this option can specify different start dates for the scheduling. Condition: The start of the operation must be within the planning horizon.

If this option is disabled, then the scheduled start date will be the earliest possible date (forward scheduling) or the latest possible start date (backward scheduling).

If the option is enabled, the "left shift" is activated and the scheduled start date is set to "now" plus planning lead time. In other words, an attempt is made to plan the operation as early as possible.



This option ignores the scheduling result, i.e. you can plan the operation before the earliest start date. If the scheduling is performed in the higher-level system, we recommend to disable this option.

**Capacity selection**

The *Capacity selection* controls for which capacity, i.e. workplace, the operation is planned that is next in the queue and must be planned next.

**The first free capacity**

The capacity that is available first in terms of time.

**Capacity having the most "capacity"**

Selects the capacity that provides the most available capacity with respect to the planning horizon.

**Capacity having the least "capacity"**

Selects the capacity that provides the least available capacity with respect to the planning horizon.

**Shortest processing time**

Selects the capacity where the processing time is the shortest (calculation includes the performance level).

Note: This is important with production variants because the values to calculate the process times can change. The capacity that is available first in terms of time.

**Shortest total setup time (static + dynamic)**

This rule for capacity selection integrates the static and dynamic setup time when assigning operations to a workplace. If this option is set, the system always assigns the operation to the workplace where the total of static and dynamic setup time is the lowest. If two machines have identical setup times, the system selects the workplace with the number that comes first in alphanumeric order.

**Shortest static setup time**

Selects the capacity where the static setup time is the shortest.

**Highest priority**

If this option is set, this rule for capacity selection uses the priority of the production variant (assigned to the relevant workplace) to assign an operation to a workplace.

If you do not use production variants, all workplaces have the same priority. The system selects the workplace that provides available capacities first. If two workplaces (production variants) have the same priority, the system selects the workplace that provides available capacities first. If two workplaces provide available capacities at the same time, the system selects the workplace with the number that comes first in alphanumeric order. If there are two production variants using different tools, the system selects the production variant using the tool with the number that comes first in alphanumeric order.

Note: The higher the priority value of a production variant, the greater the probability that this variant is used.